

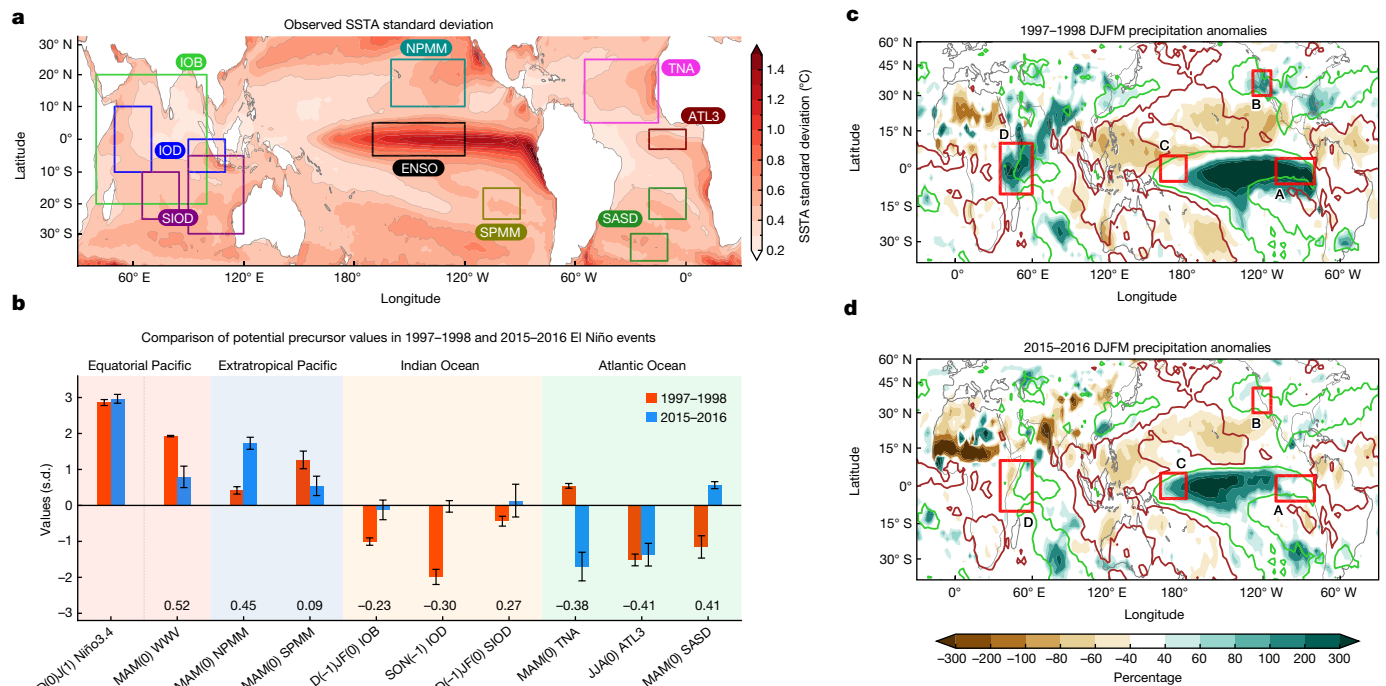


Fig. 1 | Different sources of ENSO predictability and associated different global impacts. **a**, Observed SSTA standard deviation pattern calculated from the detrended ORAS5 reanalysis during 1979–2022. The different coloured boxes represent area-averaged SSTA index regions for ENSO and other selected climate modes (Supplementary Table 1). **b**, Observed standardized Niño3.4 index and various potential precursor indices for the 1997–1998 and 2015–2016 El Niño events, with the numbers in the parentheses indicating the preceding (–1), current (0) and subsequent (1) years. The error bars show the spread (one standard deviation) among different observational products (Supplementary Table 2). The lead correlation of various indices with regard to the November–

December–January Niño3.4 index is indicated near the bottom of the plot. **c,d**, Observed precipitation anomalies (percentage) relative to climatology (shading) during 1997–1998 December–January–February–March (DJFM) (**c**) and 2015–2016 DJFM (**d**). Contours denote the significant positive (green) and negative (brown) correlations between DJFM precipitation anomalies and the DJFM Niño3.4 SSTA index that exceed the 95% confidence level, based on Student's *t*-test. The observed 1997–1998 and 2015–2016 El Niño events were associated with different precursor patterns and global climate effects, despite similar Niño3.4 index amplitudes.

and instead large amplitude NPMM warming in 2015 March–April–May. The AO SST signals are largely similar for the two events, except that the March–April–May TNA was anomalously warm in 1997 but cold in 2015. In turn, these two events evolved differently in the various basins (Supplementary Fig. 1), which lead to distinct global impacts^{31,32} (Fig. 1c,d and Supplementary Fig. 2). These two different evolutions and impacts, affected by varied precursor patterns, underscore the need to quantify the sources of prediction skill and their role in the manifestation of different SST patterns more accurately.

Recent advances have demonstrated the value of artificial intelligence (AI) in predicting ENSO with skilful forecasts at long lead time of 18–24 months (refs. 8–10). Despite emerging explainable AI methodologies¹⁰, linking the forecast skill of the AI model to specific physical processes is not yet possible, limiting our understanding of the dynamics and physical robustness underpinning the enhanced AI skill. Here we develop a low-order extended nonlinear recharge oscillator (XRO) model—which couples the ENSO recharge oscillator with autoregressive model representations for the other modes ('XRO model' section in the Methods)—to both predict ENSO events and quantify the various sources of ENSO predictability from climate mode interactions. We find that our model provides skilful and, most importantly, explainable forecasts at lead times up to 16–18 months, better than global climate models and comparable to the most skilful AI ENSO forecast model.

Efficacy boosted by climate interactions

We evaluate the XRO in simulating ENSO through a 43,000 year-long stochastically forced simulation ('Stochastically forced XRO simulations' section in the Methods) with parameter estimates derived from

1979–2022 observations (black curves in Extended Data Fig. 1). The XRO accurately simulates the fundamental observed characteristics of ENSO including its seasonal synchronization, Niño3.4 positive skewness, its interannual spectral peak, the 6–9 months lead of WWV over ENSO SST, its irregular interannual oscillations and the spring persistence barrier (Fig. 2a–d, Supplementary Text 1 and Supplementary Figs. 3 and 4). The XRO also accurately reproduces the observed seasonal characteristics of the other climate modes including their seasonal synchronizations and autocorrelations (Supplementary Figs. 5 and 6). In addition, the XRO realistically simulates the observed lead–lag relationships between ENSO and all the other climate modes with the range of XRO realization cross-correlations encompassing the observations (Fig. 2e–i). Simulating these observed relationships is a major challenge for climate models (Supplementary Fig. 7).

Next, we perform out-of-sample XRO reforecasts by fitting the model for 1950–1999 (50 years) and verifying it independently for the 2002–2022 period ('Out-of-sample reforecasts' section in the Methods). The correlation skills of the Niño3.4 reforecasts are compared with a nonlinear recharge oscillator model (nRO), the real-time International Research Institute for Climate and Society (IRI) operational models, and the most skilful AI ENSO forecast model^{8,9} (Fig. 2m). The skill of the simple nRO is comparable with the ensemble mean of the IRI statistical models. With mode interactions considered, the XRO outperforms the ensemble mean of the IRI dynamical models at long lead times (greater than 9 months) with skill scores comparable to the AI model. We also test the model by verifying the early period (1950–1970) and the middle period (1972–1992) independently. The XRO outperforms the nRO regardless of which of the verification periods is used to assess the skill (Extended Data Fig. 2), suggesting the importance of mode